Research paper

One of the in demand scientific questions in the world of computer vision, "Object Detection" (finding a person in a photo) and "Single-Camera Tracking" (following a person as they walk across one screen) are **largely solved problems**. However, a significant gap exists when a person leaves the view of one camera and enters another. This is known as Person Re-Identification (Re-ID).

In real-world environments such as retail stores, transportation hubs, and public buildings, humans solve this easily by using common sense—we know a person can't teleport across a building in one second. However, most AI models are still incapable of understanding this basic environmental context as humans do. They treat every camera view as a brand-new problem, relying entirely on what a person is wearing to identify them. If two people wear similar blue jackets, the AI often gets confused.

Current research focuses heavily on "Deep Learning" to recognize colors and textures (Zheng et al., 2015). While these models are powerful, they are "static"—they ignore the physical reality of the environment. Existing high-end solutions that do use "logic" (like tracking someone's exact walking speed) often require massive computing power or raw video files that are too large to handle. There is a need for a "middle ground": a lightweight way to add time-and-space rules to standard photo-based matching.

Importance and value solving is vital for "Smart Cities" and retail. For example, in a large airport, security needs to know if a specific bag-owner moved from the check-in desk to the gate. Or for small-medium sized retail stores who cannot afford to build whole infrastructure, but wants to understand the customer journey in their store. If the system relies only on clothes, it fails in crowded areas. Improving this through "logic" rather than just "better cameras" makes these systems more reliable and cheaper to run.

Research Question:
To what extent can adding simple rules about time and camera locations improve the accuracy of identifying people across multiple cameras, compared to models that only look at physical appearance?

Methodology

Fundamentally, re-identification tasks can be framed as a **matching problem**. Given an image of a person captured by camera A, the objective is to identify the same individual in images captured by cameras B, C, or D.

Modern re-identification systems typically follow a three-stage pipeline.

1. Person Detection

A continuous video stream is first divided into frames, and each frame is passed through a deep learning algorithm that detects the location of each person in the image. These algorithms generate **bounding boxes** that isolate individual people.

Common algorithms used for this task include:

- YOLO
- R-CNN
- Fast R-CNN

These detection models identify where people are located in the frame so that each person can be processed separately.

2. Feature Extraction

Each cropped image of a detected person is then passed through a **convolutional neural network (CNN)** that converts the image into a numerical vector representation.

This process can be summarized as:

Person image → CNN → numerical vector

Typically, this vector has around **2048 dimensions** depending on the architecture used. The neural network learns to encode visual attributes such as clothing color, body shape, accessories, and other distinguishing patterns. These vectors are commonly referred to as **embeddings** or **feature representations**. For example:

Person A → [0.12, 0.98, 0.43, ..., 0.51]

3. Similarity Matching

Once images are encoded into vectors, the system compares vectors from different cameras. This comparison typically involves computing a distance or similarity measure between embeddings.

Common measures include:

- cosine similarity
- Euclidean distance

If the vectors are very similar, the system predicts that the images correspond to the same person.

Current research in re-identification problems focuses primarily on improving the **feature extraction stage**. Researchers attempt to design better neural network architectures such as Vision Transformers, ResNet, or EfficientNet that capture more visual information. These architectures are often combined with advanced training techniques and data augmentation strategies to improve embedding quality.

However, the fundamental assumption remains unchanged: **identity is determined primarily through visual characteristics of the image**.

This project explores whether incorporating **spatial and temporal context** can improve identification performance.

To test this hypothesis, I will compare two models:

1. a baseline appearance-only model
2. a context-enhanced model that incorporates environmental information

Importantly, I will **not attempt to reconstruct or train deep neural networks from scratch**. Instead, I will use pretrained neural networks to extract embeddings and then use traditional machine learning models for classification.

The proposed pipeline is as follows.

Step 1: Extract environmental information

Metadata will be extracted from image filenames, including:

- frame number (proxy for time)
- camera identifier (proxy for spatial location)

Step 2: Extract embeddings

A pretrained neural network will be used to extract embeddings from each image.

Step 3: Construct pairwise observations

Pairs of images will be constructed in order to transform the problem into a classification task.

Pairs will be defined as follows:

- **Positive pairs:** images of the same individual captured by different cameras
- **Negative pairs:** images of different individuals

For each pair, the following features will be computed:

- cosine similarity between embeddings
- time difference between observations
- camera pair identifier
- travel feasibility indicator (in later stages)

These variables will form the input features for the classification models.

Step 4: Train classification models

Two models will be estimated.

Baseline Model

This model will rely only on visual appearance.

Input feature:

- cosine similarity

Model:

- logistic regression

This model will serve as a benchmark for evaluating how well visual appearance alone can identify individuals across cameras.

Context-Enhanced Model

This model will include contextual information in addition to visual similarity.

Input features:

- cosine similarity
- time difference between frames
- camera pair indicator

Two models will be evaluated using standard classification metrics:

- Accuracy
- Precision
- Recall
- F1 score

In later stages of the project, performance will also be evaluated using standard re-identification metrics such as **Rank-1 accuracy** and **mean Average Precision (mAP)**.

The expectation is that the **context-enhanced model will outperform the appearance-only baseline**.

Data Review

The raw image filenames were parsed to extract metadata describing each observation. After preprocessing, the resulting dataset contained **12,936 observations** and **six variables**, including image path, person identifier, camera identifier, sequence identifier, and frame number.

Summary Statistics

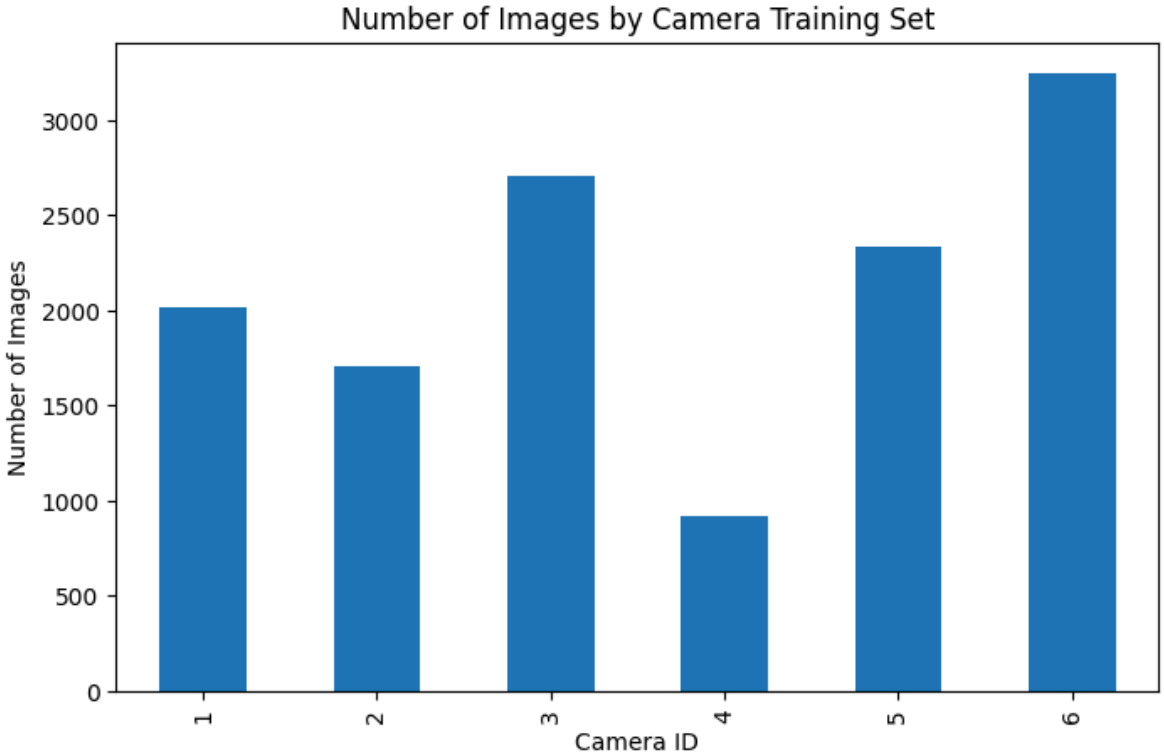
The dataset includes:

- 12,936 images
- 751 unique identities
- 6 camera locations

The distribution of images per individual shows that most individuals appear between 9 and 22 times, with a median of approximately 15 images per person.

Statistic	Images per Person
Mean	17.23
Median	15
Min	2
Max	72

Distribution of Images per Person Training Set
Frequency
Number of Images
0
20
40
60
80
100
0
10
20
30
40
50
60
70


Number of Images by Camera Training Set
Number of Images
Camera ID
0
500
1000
1500
2000
2500
3000
1
2
3
4
5
6

Images are relatively evenly distributed across the six cameras, ensuring that no single camera dominates the dataset. This is important for evaluating cross-camera identification performance.

A key requirement for re-identification tasks is that individuals appear in multiple camera views. The dataset satisfies this condition.

The number of cameras per identity has the following distribution:

Statistic	Cameras per Identity
Mean	4.34
Median	5
Min	2
Max	6

This indicates that most individuals appear in **four to five cameras**, providing sufficient cross-camera observations for training and evaluation.

Anticipated Challenges

First, the dataset contains **cropped images rather than continuous video**, meaning that frame numbers provide only an approximate measure of time. As a result, temporal constraints must be interpreted cautiously.

Second, appearance-based embeddings extracted from ImageNet-trained networks are not optimized specifically for re-identification tasks. This may limit the performance of the baseline model.

Third, the number of potential image pairs grows rapidly as the dataset expands. For our 13,000 images, full pairwise matching would generate over 169 million pairs—a scale that is computationally inefficient. More importantly, this approach creates a massive class imbalance between positive and negative pairs. To address this, we use a negative sampling strategy, pairing each positive sample with 5–6 negative ones.

Results & Challenges

The first step of the analysis involved identifying an appropriate baseline model for extracting visual embeddings in the person re-identification task. Several pre-trained models (three in total) were evaluated with the goal of selecting the one that produces the most discriminative feature representations.

To ensure a fast and interpretable comparison, a simple metric was employed: the **average cosine similarity** between image pairs. Specifically, cosine similarity was computed separately for:

- **Positive pairs** (images of the same identity), and
- **Negative pairs** (images of different identities).

Cosine similarity is chosen over Euclidean distance because deep embedding spaces encode identity primarily in the **direction of the feature vector rather than its magnitude**. Factors such as lighting conditions, camera angle, and background can affect vector magnitude, while the directional structure remains more stable. As a result, cosine similarity provides a more robust and semantically meaningful measure of identity similarity in this context.

An effective embedding model is expected to produce **high similarity scores for positive pairs** and **low similarity scores for negative pairs**, thereby maximizing the separation between the two distributions.

Across the evaluated models, the **vanilla RenNet50 model** demonstrated the strongest discriminative performance, exhibiting the largest gap between average cosine similarity of positive and negative pairs. Based on this empirical evidence, it was selected as the primary embedding extractor for subsequent experiments.

Model	**Positive Mean Similarity**	**Negative Mean Similarity**	**Separation (Pos - Neg)**
Vanilla ResNet50 (ImageNet)	**0.6231**	**0.5543**	**0.0688**
torchreid ResNet50	0.7875	0.752	0.0355

OSNet x1.0 (Market-1501 Re-ID)	0.6696	0.6124	0.0572

To test whether environmental information can improve identification performance, a context-enhanced model was constructed by augmenting the baseline with temporal and spatial features.

For each pair of images, the following additional variables were included:

- **Time difference**: computed as the absolute difference in frame numbers (used as a proxy for elapsed time)
- **Camera pair indicator**: a categorical variable representing transitions between camera locations

Importantly, no strict deterministic constraints (such as maximum allowable travel time between cameras) were imposed. Instead, temporal and spatial variables were introduced as **soft features**, allowing the model to learn their relevance directly from the data. This design choice reflects uncertainty in the data, as frame numbers only approximate real time and movement dynamics between cameras are not explicitly observed.

Model performance was evaluated using both **classification metrics** and **re-identification metrics**.

Classification metrics included:

- Accuracy
- Precision
- Recall
- F1-score

These metrics assess how well the model distinguishes between matching and non-matching pairs.

However, the primary evaluation relies on **re-identification metrics**, which better reflect the practical objective of the task:

- **Rank-1 accuracy**: the proportion of cases where the correct identity is the top-ranked match
- **Top-5 accuracy**: the proportion of cases where the correct identity appears among the top five candidates

These metrics are standard in re-identification literature and directly measure retrieval performance.

The main results of the analysis are summarized in Table below

Model	**Rank-1**	**Top-5**
Baseline (appearance only)	0.10	0.20
Context (+ time, + space)	0.03	0.09

The baseline model achieves a Rank-1 accuracy of approximately 0.10 and Top-5 accuracy of approximately 0.21. In contrast, the context-enhanced model achieves substantially lower performance, with Rank-1 accuracy around 0.03 and Top-5 accuracy around 0.09.

Contrary to the initial hypothesis, the inclusion of temporal and spatial context **significantly degrades performance**. The context-enhanced model performs worse than the baseline across all re-identification metrics.

This result is particularly striking given that contextual information was expected to provide additional constraints and improve matching accuracy.

Several factors may explain this unexpected outcome.

First, the temporal variable is inherently noisy. Frame numbers serve only as a proxy for time and do not capture true time between observations. As a result, the model may learn misleading temporal relationships that do not correspond to real-world movement.

Second, the model does not enforce physical constraints on movement between cameras. Without explicit rules (such as minimum travel time), the model treats temporal and spatial features as generic inputs, allowing implausible transitions to influence predictions.

The results suggest that contextual information is not inherently beneficial in re-identification tasks. Instead, its effectiveness depends on how it is modeled and integrated into the system.

Naively adding temporal and spatial features as independent variables may introduce noise and reduce the quality of similarity estimation. More sophisticated approaches—such as enforcing physical constraints, learning context jointly with embeddings, or modeling transitions probabilistically—may be required to fully leverage contextual information.

This analysis demonstrates that, despite intuitive appeal, simple incorporation of time and space information does not improve—and can significantly harm—re-identification performance. This

highlights the importance of careful feature design and alignment between model structure and task objectives.

Next steps

Given the performance deterioration with basic contextual information, the next approach towards improvement might include the following:

1. Feasibility-Based Temporal Constraints

Feasibility metric based on the empirical distribution of time differences observed in positive pairs.

Instead of treating time difference as a raw continuous variable, the model can learn what constitutes a *plausible transition* between cameras. Specifically:

- For each camera pair, the distribution of time differences among positive matches can be estimated
- This distribution can then be used to compute a **likelihood or feasibility score** for any candidate pair

For example:

- If two images appear in cameras that typically require at least 30 seconds of travel time, a near-zero time difference should be penalized
- Conversely, time differences that fall within typical ranges should be assigned higher feasibility

2. Normalization of Temporal Features

Another limitation of the current approach is the use of log time differences, which may vary significantly across camera pairs and sequences.

To address this, time variables can be **normalized**:

- Standardization (z-scores) within each camera pair
- Scaling relative to typical transition times

Normalization ensures that temporal features are **comparable across different contexts** and prevents them from disproportionately influencing the model due to scale differences.

3. Interaction Effects Between Appearance and Context

In the current specification, cosine similarity, time difference, and camera indicators enter the model independently. However, the relevance of contextual information likely depends on visual similarity.

For instance:

- When cosine similarity is high, temporal constraints may be less important
- When cosine similarity is low, contextual feasibility may play a stronger role

This suggests the inclusion of **interaction terms**, such as:

- cosine similarity × time difference
- cosine similarity × camera pair

These interaction effects allow the model to capture **conditional relationships**, where context influences decisions differently depending on visual similarity.

4. Isolating the Effects of Time and Space (the most important)

To better understand the source of performance degradation, future experiments should evaluate temporal and spatial features separately.

Specifically:

- A model including only **time difference** (without camera information)
- A model including only **camera pair indicators** (without time)

This ablation-style analysis would help determine whether:

- Temporal noise is the primary driver of performance decline
- Spatial information alone provides useful signal
- Or whether both features contribute negatively when combined

Such analysis is critical for disentangling the effects of different contextual variables.